

Epic 1: Model Selection and Architecture

1. Rationale for Gemini 3.8 Flash Selection

The EduGenie core pipeline relies heavily on the gemini-3.8-flash model for processing complex language tasks. This specific model was chosen to fulfill production demands across four distinct vectors: low operational latency, cost efficiency, structured prompting compliance, and massive context boundaries.

Tasks like generating comprehensive, multi-tiered learning pathways, dynamically constructing structured, multi-choice quizzes with built-in explanation rationales, and compiling concise text summaries require an engine that understands complex contextual formatting directions. The Gemini Flash tier delivers rapid processing times that are crucial for interactive web interfaces. This prevents web gateway timeouts without sacrificing the logical accuracy needed for educational tasks.

2. Architectural Design Patterns

The platform relies on a hybrid architectural blueprint that blends cloud computing with local execution capabilities. This structural separation isolates concerns, optimizes resource utilization, and maintains data flexibility:

Asynchronous Web Gateway Layer: Built entirely on FastAPI, the web framework manages concurrent user connections using non-blocking I/O structures. This setup routes independent data flows smoothly without locking execution threads during intensive AI inference tasks.

State-Free Functional Micro-Modules: The business logic is organized into decoupled Python files. Each script functions as an independent service layer that handles a single specific task type. This modular structure simplifies code maintenance, speeds up unit testing, and makes it easy to swap model checkpoints without modifying the core web app code.

Hybrid Cloud-Edge Model Infrastructure: The architecture splits its processing workloads based on resource requirements. General tasks are sent to cloud-hosted endpoints via secure

SDK wrappers. Meanwhile, specialized concept explanations run on a local CPU using a fine-tuned Hugging Face transformer model pipeline. This hybrid setup ensures the system remains resilient and available even if cloud network limits are exceeded.